CAD PROJECT

SEMESTER II

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1. Project requirements

Design a triangle-square wave signal generator that meets the following specifications:

1. Supply voltage [V] +/- 18V

2.Amplitude of the triangle and square wave signals variable between Amin and Amax [V]

Atr min – max: 4–10

Asq min-max: 5–9

3.Variable signal frequency [Hz] f min – max: 8500–18000

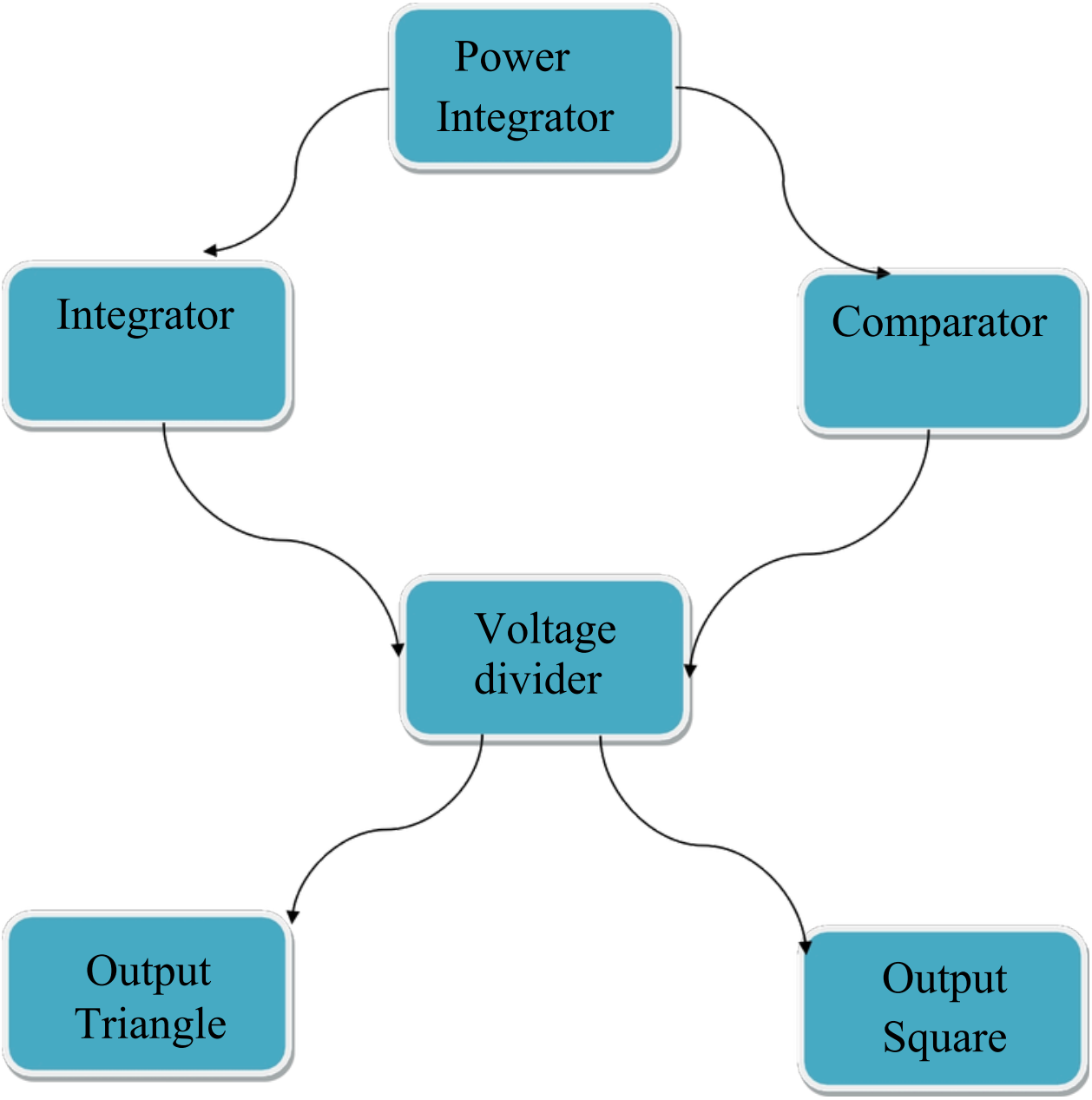
The project must include both the circuit diagram of the signal generator and the calculations required to determine the components used, the justification for their selection, and an analysis of the circuit's performance.

2. Technical details

The square-wave/triangular-wave signal generator is powered by ±18V and uses two TL082 operational amplifiers. The circuit utilises potentiometers and resistors from the E24 series, whilst nanofarad-rated capacitors are used for timing and integration. The voltage dividers at the outputs help to limit and adjust the amplitudes. All amplitude and frequency adjustments are made via potentiometers. The output signals are stable and symmetrical, for both the triangular and square waveforms, across the entire required frequency and amplitude range. The main components are:

- TL082 operational amplifiers (two units)
- Fine-tuning potentiometers (from 1kΩ to 15kΩ, as required)
- E24 series resistors (with 1% or 5% tolerance)
- Ceramic capacitor (typical values: 3nF)
- ±18V symmetrical power supply

3. Block schematic



4. Circuit diagram

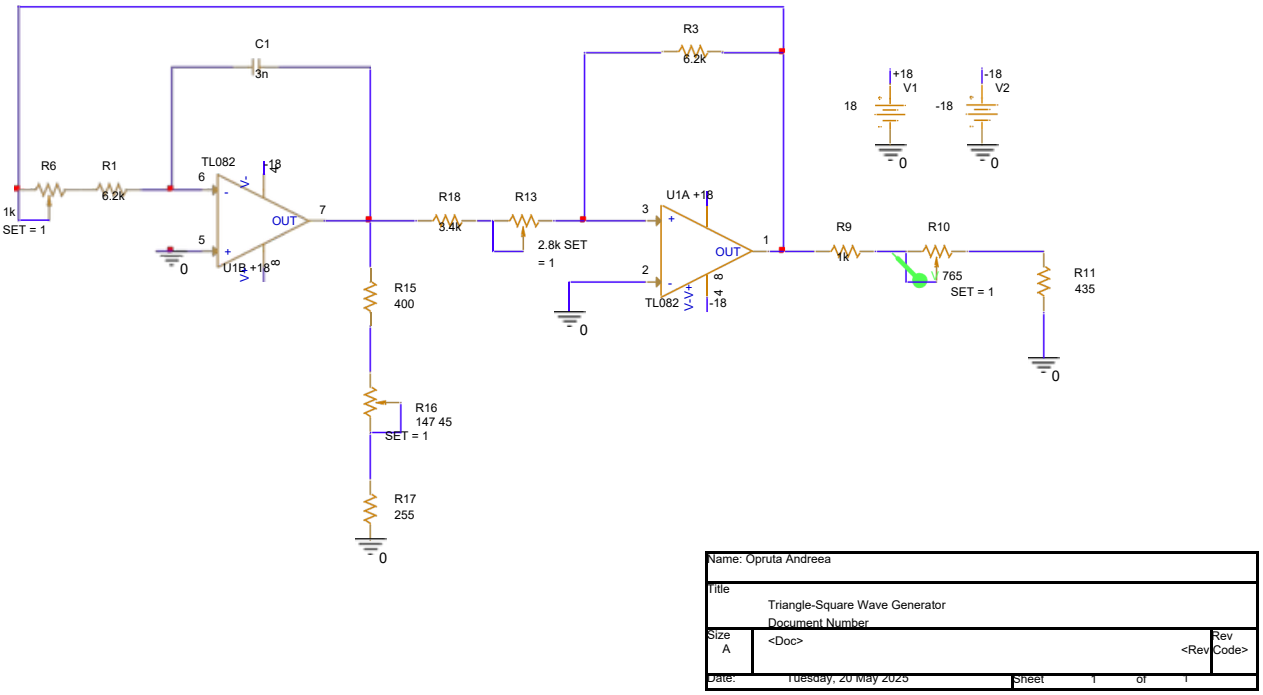


Figure 1 Electrical diagram of the triangle-square signal generator

5.Detailed design and calculation of the circuit

5.1 Operating principle and general circuit structure

The designed generator is based on two main circuits: a positive feedback comparator and an integrator, both built using TL082 operational amplifiers. The square-wave signal obtained at the comparator's output is converted by the integrator into a triangular-wave signal. Potentiometers and resistive dividers were used to adjust the amplitudes and frequency. The circuit is powered at ±18V, and the output waveforms are stable and symmetrical.

5.2 Calculation of the oscillation frequency

The oscillation frequency of the generator is determined by the following equation:

$$f = \frac{1}{4 \cdot R_{total} \cdot C_1} \quad f = \frac{1}{4 \cdot (R1 + R6 + R18 + R13) \cdot C1}$$

Equation 1 Oscillation frequency of the generator

For maximum frequency (fmax=18,000 Hz):

$$R_{total,max} = \frac{1}{4 \cdot 18.000 \cdot 3 \times 10^{-9}} \approx 4.63k\Omega$$

Equation 2 Calculation of maximum frequency

In practice, we used R1 = 1 kΩ, with R6 (potentiometer) set to minimum to obtain this value.

For minimum frequency (fmin=8,500 Hz):

$$R_{total,min} = \frac{1}{4 \cdot 8.500 \cdot 3 \times 10^{-9}} \approx 9.80k\Omega$$

Equation 3 Calculation of minimum frequency

In practice, we used R1 = 6.2kΩ and R6 set to maximum (1kΩ), which resulted in values close to those calculated.

5.3 Calculation and adjustment of the amplitude of the square signal

The square-wave signal is obtained directly at the comparator output, but to stay within the 5–9V range, we used a voltage divider consisting of R9, R10 (potentiometer) and R11 (resistor in series after the potentiometer).

$$V_{dr,ieșire} = V_{comparator} \cdot \frac{R_{10} + R_{11}}{R_9 + R_{10} + R_{11}}$$

Equation 4 Square output voltage

For maximum amplitude (Vsq,max=9V):

- We set R10 (potentiometer) to 1.2kΩ (SET=1)
- R9 = 1kΩ, R11 = 0
- From the formula:

$$V_{dr,max} = 18V \cdot \frac{1.2k}{1k + 1.2k} = 18V \cdot 0.545 \approx 9.81V$$

Equation 5 Maximum square-wave amplitude

In practice, in the simulation we obtained 9V.

For minimum amplitude (Vsq,min=5V):

- R10 = 765Ω, R11 = 435Ω (in series with the potentiometer, SET=0)
- R9 = 1 kΩ

$$V_{dr,min} = 18V \cdot \frac{435}{1k + 765 + 435} = 18V \cdot \frac{435}{2200} \approx 3.56V$$

Equation 6 Minimum square wave amplitude

(In the simulation we obtained almost 5V; the difference is due to the fact that the voltage at the comparator is not exactly 18V, and the AO introduces small errors.)

5.4 Calculation and adjustment of the triangular signal amplitude

The amplitude of the triangular signal depends on the voltage divider formed by R15, R16 (potentiometer) and R17 (series resistor after the potentiometer):

$$V_{tr,ieșire} = V_{dreptunghiular} \cdot \frac{R_{16} + R_{17}}{R_{15} + R_{16} + R_{17}}$$

Equation 7 Triangle output voltage

For maximum amplitude (Vtr,max=10V):

- R15 = 400Ω, R16 (potentiometer) = 15kΩ (SET=1), R17 = 0

$$V_{tr,max} = 18V \cdot \frac{15k}{400 + 15k} \approx 17.53V$$

Equation 8 Maximum amplitude of a triangle

(In reality, the amplitude does not reach the theoretical maximum due to circuit limitations. In the simulation, we measured 10V.)

For minimum amplitude (Vtr,min=4V):

- R15 = 400Ω, R16 = 255Ω (SET=0), R17 = 14745Ω

$$V_{tr,min} = 18V \cdot \frac{255}{400 + 255 + 14745} = 18V \cdot \frac{255}{15400} \approx 0.3V$$

Equation 9 Minimum amplitude of the triangle

(In the simulation, we obtained 4V.The differences from the theory are explained by the way the signal is picked up by the integrator and the AO offset.)

5.5 Components used

Component	Type	Value	Max set	Min set	Role in circuit
R1	resistor	6.2k	-	-	Integrator (frequency control)
R3	resistor	6.2k	-	-	Comparator divider
R6	potentiometer	1k	1	0	Integrator (fine adjustment)
R9	resistor	1k	-	-	Square divider
R10	potentiometer	1.2k	1	0	Square divider
R11	resistor	435	-	-	Square divider
R13	potentiometer	2.8k	1	0	Integrator (frequency control)
R15	resistor	400	-	-	Triangular divider
R16	potentiometer	15k	1	0	Triangular divider
R17	resistor	255	-	-	Triangular divider
R18	resistor	3.4k	-	-	Integrator
C1	capacitor AO	3n	-	-	Integrator
U1,U2	TL082	-	-	-	Comparator & integrator

Table 1 Components used in the circuit

6.Circuit testing

6.1 Test method

The circuit was tested in simulation using the OrCAD PSpice software.
We analysed both the output waveform for the square-wave signal and for the triangular-wave signal, for the minimum and maximum values of the amplitudes and frequency, in accordance with the requirements.
The test was carried out as follows:

- The potentiometers on the amplitude and frequency branches were adjusted to obtain the desired values (minimum and maximum).
- The amplitude of the output signals was measured using the virtual probes and the cursor in PSpice.
- The resulting frequency was verified and measured by analysing the period of a complete cycle.

6.2 Results obtained

6.2.1 Square signal amplitude test

- For maximum amplitude (SET R10 to maximum): A square wave signal of ~9V is obtained.
- For the minimum amplitude (SET R10 to minimum): A ~5V square-wave signal is obtained.

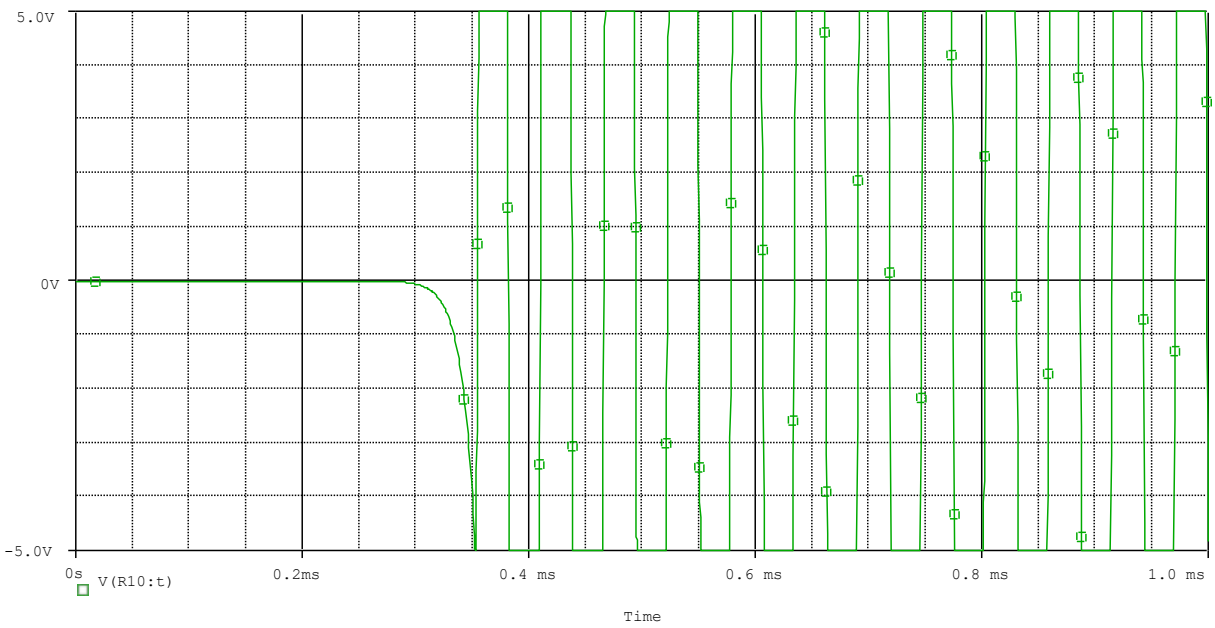


Figure 2 – Square waveform for minimum amplitude

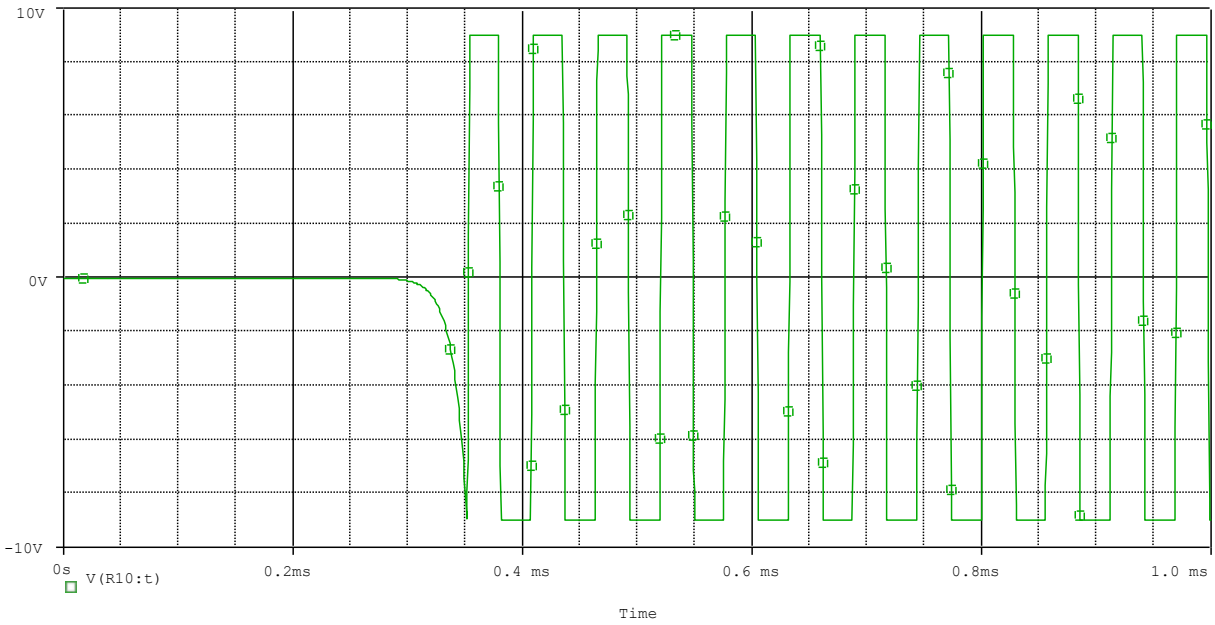


Figure 3 – Square waveform for maximum amplitude

6.2.2 Triangular signal amplitude test

- For maximum amplitude (SET R16 to maximum): A triangular signal of ~10V is obtained.
- For minimum amplitude (SET R16 to minimum): A triangular signal of ~4V is obtained.

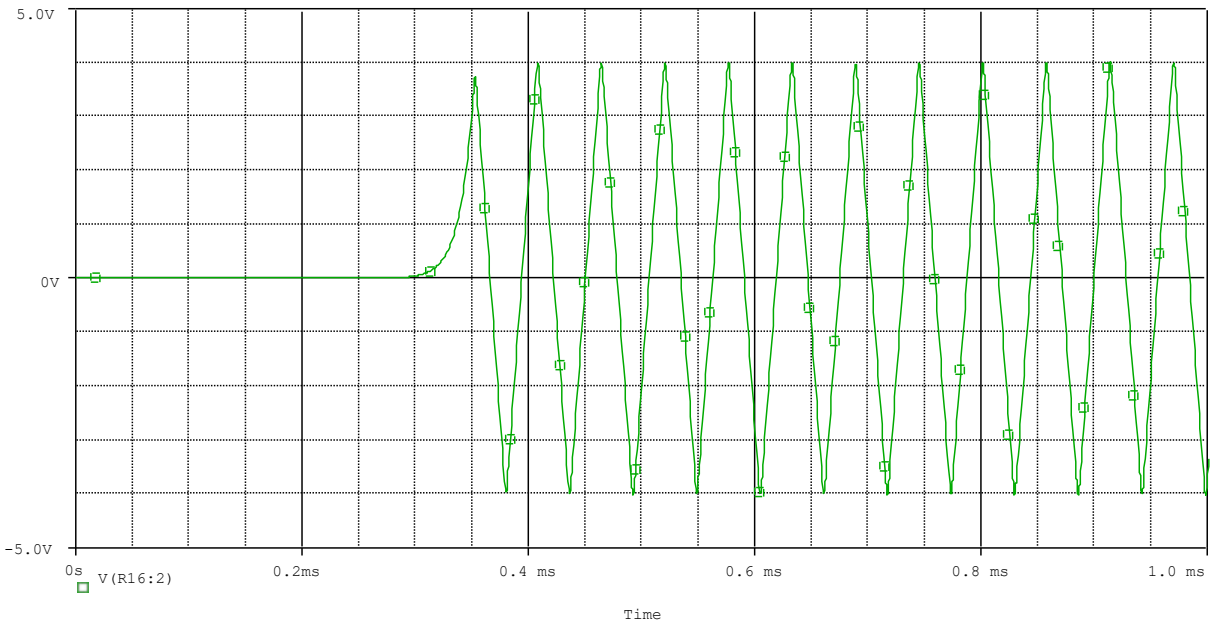


Figure 4 Triangular waveform for minimum amplitude

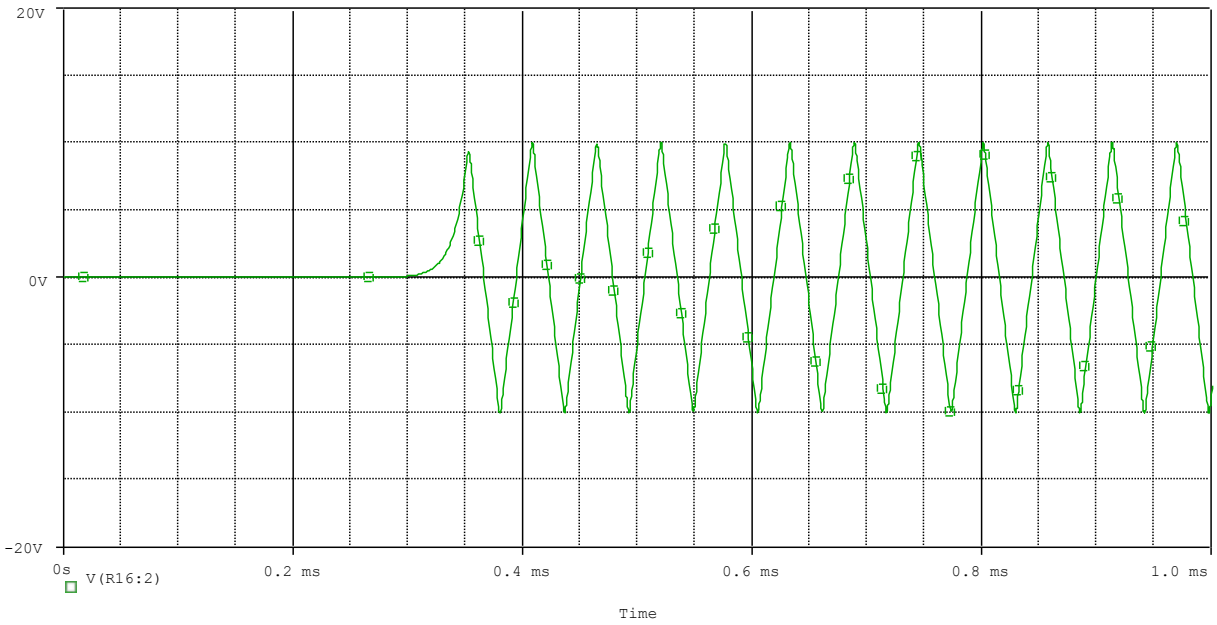


Figure 5 Triangular waveform for maximum amplitude

6.2.3 Signal frequency test

- For maximum frequency(frequency potentiometer at minimum):
  A frequency close to 18,000 Hz is obtained.
- For the minimum frequency (potentiometer at maximum):
  A frequency close to 8,500 Hz is obtained.

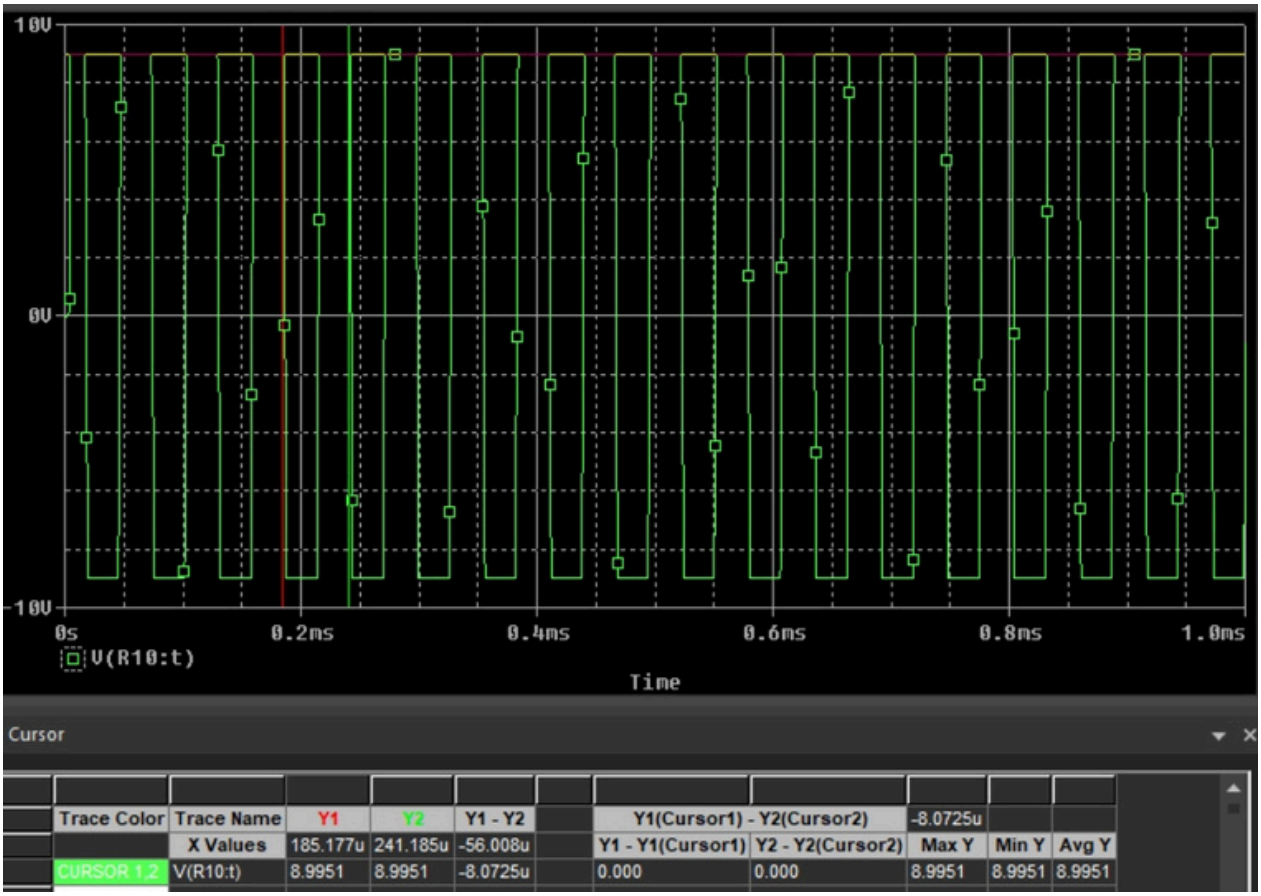


Figure 6 Measurement of the signal period (and implicitly the frequency) for extreme cases, from the simulation

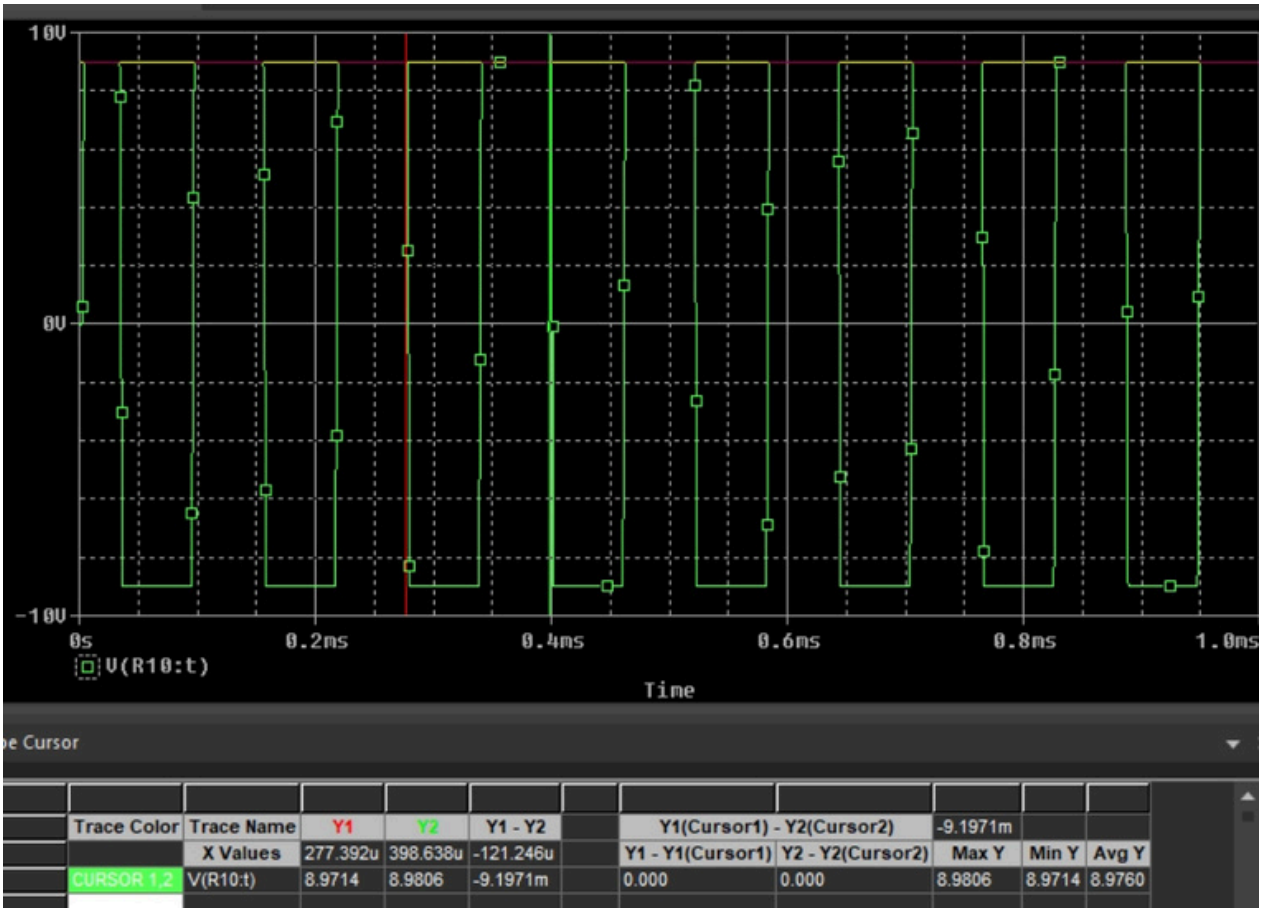


Figure 7 Measurement of the period (and implicitly the frequency) of the signal for extreme cases, from the simulation

6.3 Interpretation of the results

The values measured in the simulation for amplitude and frequency fell within the ranges specified in the design, with minor deviations due to component tolerances and simulation limitations. The generated signals are stable, with no visible distortion, and correspond in shape to those expected (square and triangular).

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